

Solution Requirements

Date	30 June 2026
Team ID	
Project Name	AI and Machine Learning Project
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users and bank administrators log in securely using username and password authentication.	High
2	Authorization levels	Applicants can submit applications and view results, while admins can manage applications and view reports.	High
3	External interfaces	The system connects with credit score databases and external banking systems if required.	Medium

4	Transaction processing	The system receives applicant details, processes the information, and generates approval or rejection predictions.	High
5	Reporting	Generate reports showing total applications, approved applications, rejected applications, and prediction accuracy.	Medium
6	Business rules, etc.	Applicants with higher credit scores and stable income have higher approval chances. Missing information leads to rejection or manual review.	High
7	Compliance to laws or regulations	The system must protect applicant personal and financial information, ensure secure storage of data, and comply with banking data privacy regulations and data protection standards such as General Data Protection Regulation principles and local banking policies.	High
8	<i>Other</i>	The system should allow administrators to retrain or update the machine learning model using new data.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The system should provide predictions quickly after data submission.	Prediction generated within 2 seconds
2	Scalability	The system should support increasing numbers of applications and records.	Supports more than 10,000 applications
3	Security & Data Privacy	Applicant data should be encrypted and protected from unauthorized access.	AES-256 encryption and secure login
4	Reliability & Availability	The system should remain available for users with minimal downtime.	99.9% uptime per month
5	Usability & Accessibility	The interface should be simple and easy to use for both applicants and administrators.	User-friendly interface with clear navigation
6	<i>Other</i>	The system should be easy to maintain and update in the future.	Modular and maintainable code structure